

# LIMIN WANG, PhD

Phone: (631) 882-0228 | Email: [liminwang636@gmail.com](mailto:liminwang636@gmail.com) | [Linkedin](#) | [Github](#) | U.S. Citizen

## SUMMARY

Data Scientist with 3+ years of experience applying machine learning and advanced analytics to optimize manufacturing and energy operations. Specializes in anomaly detection, predictive modeling, and data-driven decision-making to enhance efficiency, mitigate risks, and drive business impact.

## SKILLS

Python, SQL, SK-Learn, Pytorch, Scipy, Numpy, Jupyter, Google Colab, Pandas, Spark, Power BI, Matplotlib, Seaborn, Streamlit, Flask, Beautiful Soup, Time Seires Forecasting, Git, Github, Linux, Parallel Computing, Matlab, C++, Fortran, AWS, Docker

## EXPERIENCE

|                              |                    |
|------------------------------|--------------------|
| <b>Techlent Inc.</b>         | Remote             |
| <i>Data Scientist Fellow</i> | Aug 2022 - Present |

- **Loan Default Detector**

- To accurately estimate the default risk of applicants and help loan officers in decision making, built a Loan Default Detector based on binary classification models.
- Processed and analyzed historical loan data; utilized multiple classification models, including Logistic Regression, Random Forest, and XGBoost; achieved an AUC-ROC score of 80% with XGBoost.
- Wrapped the best-performing model as a Flask APP and developed it on AWS.

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|-------------------------------------------------|---------------------|
| <b>Rutgers University</b>                       | Piscataway, NJ      |
| <i>Senior Research Scientist/Data Scientist</i> | Aug 2019 - Aug 2022 |

- **Predictor of Novel Solar Panel Materials**

- To accelerate the innovation process of next-generation solar panels with better energy harvesting efficiency and lower the R&D expense, developed a Material Candidate Predictor based on binary classification models.
- Conducted feature engineering and feature selection based on domain knowledge; experimented with multiple classification models (Logistic Regression, Random Forest, and XGBoost).
- Achieved a precision score of 81% with the best performing model (XGBoost), reducing R&D expenses by 55%, and accelerating material discovery time by 70%. Used SHAP values to interpret model outputs.

- **Bandgap Engineering in Novel Topological Quantum Materials**

- Optimized ZnSnO3 bandgap through engineered polarization and topological Berry's Phase, enhancing its electronic properties.
- Identified ZnSnO3 as a strong candidate for energy-efficient field-effect transistors (FETs), supporting low-power semiconductor applications.

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|-------------------------------|---------------------|
| <b>University of Maryland</b> | College park, MD    |
| <i>Research Scientist</i>     | Aug 2014 - Jul 2019 |

- **Model-Guided Design of Novel Topological Quantum Materials**

- Developed computational models to predict non-trivial topological materials (RbSb3, BaAl4, Pt3C12S12) using Material Calculations and Machine Learning models.
- Collaborated with experimentalists to optimize material synthesis and validate theoretical predictions, accelerating material development for next-gen electronic devices.
- Built an automated Python pipeline to analyze and visualize computational results, reducing manual processing time and increasing research efficiency.

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|---------------------------------------|---------------------|
| <b>Brookhaven National Laboratory</b> | Upton, NY           |
| <i>Research Associate Physics</i>     | Jan 2011 - Dec 2013 |

- **Computational Modeling of Quantum Superconducting Materials**

- Conducted first-principles simulations to analyze the electronic structure of topological superconductors, improving understanding of quantum transport properties.
- Co-developed Orbital-Symmetry Wannier (OSW), a Fortran and C++-based software package for evaluating impurity effects in superconducting materials, enabling more accurate material property predictions.

## PROJECT

### Financial Market Predictor

- To aid investors, financial analysts, and policymakers in making informed decisions, build a Financial Market Predictor via Time Series Forecasting.
- Collected and processed historical NASDAQ data; utilized ARIMA, Double Exponential Smoothing, Decision Tree, and Gradient Boost; achieved a Max Absolute Error Percentage of 2.01% with Gradient Boost.

## EDUCATION

### Nanjing University

*Ph.D. in Condensed Matter Physics*

Nanjing, China

Sep 2004 - Dec 2008

## CERTIFICATES

- AWS Certified Developer – Associate 2022
- AWS Certified Solutions Architect – Associate 2023

## Selected Publications

1. **Limin Wang**, Karin M. Rabe, *Discovery of transparent conductors using machine learning*, (prepared).
2. **Limin Wang**, Karin M. Rabe, *Berry's phase and band gap engineering via ferroic distortion: cubic ZnSnO<sub>3</sub>*, (to be submitted).
3. **Limin Wang**, Karin M. Rabe, *First principles investigation of effects of Nb-substitution in EuTiO<sub>3</sub>*, (submitted).
4. K. F. Wang, R. Mori, Z. Wang, **Limin Wang**, J. H. Son Ma, D. W. Latzke, D. Graf, J. D. Denlinger, D. Campbell, B. Bernevig, A. Lanzara, and J. Paglione, *Crystalline symmetry-protected non-trivial topology in prototype compound BaAl<sub>4</sub>*, **npj Quantum Mater.** **6**, 28 (2021)
5. Y. Nakajima, R. Wang, **Limin Wang**, S.R. Saha, I-L. Liu, Y.S. Eo, D. Graf, Z. Liu, S.V. Borisenko, P.Y. Zavalij, J. Paglione, *Planckian dissipation and scale invariance in a quantum-critical disordered pnictide*, **Communications Physics** **3**, 181(2020).
6. R. Stillwell, **Limin Wang**, X.F. Wang, D. Campbell, J Paglione, J.R. Jeffries, *Observation of double-collapse transitions in CaRbFe<sub>4</sub>As<sub>4</sub> layered superconductor*, **Physical Review B** **100**, 045152 (2019).
7. Igor Zaliznyak, David Wendt, Emil Bozin, Joerg Neuefeind, Katharine Page, Wei Ku, **Limin Wang**, B Fultz, Alexei Tkachenko, *Entropic elasticity and negative thermal expansion in a simple cubic crystal*, **Sci. Adv.** **5**, eaay2748 (2019)
8. Hyunsoo Kim, Steven Ziemak, Paul Syers, **Limin Wang**, Halyna Hodovanets, Kefeng Wang, Rongwei Hu, Yasuyuki Nakajima, Johnpierre Paglione, Makariy A. Tanatar, Ruslan Prozorov, Philip Brydon, Daniel Agterberg, Jonathan D. Denlinger, *Nodal Topological Superconductivity in Noncentrosymmetric YPtBi*, **Sci. Adv.** **4**: eaao4513(2018)
9. P. M. R. Brydon, **Limin Wang**, M. Weinert, and D. F. Agterberg, *Pairing of  $j = 3/2$  fermions in half-Heusler superconductors*, **Physical Review Letter** **116**, 177001 (2016)
10. Wei Ku, Tom Berlijn, **Limin Wang**, Chi-Cheng Lee, and Chia-Hui Lin, Chapter 7: *First-Principles Studies in Fe-Based Superconductors*, (Iron-Based Superconductivity, Springer, New York, 2014).
11. **Limin Wang**, Shaocong Shi and Weiyi Zhang, Anisotropic localization behavior of graphene in the presence of diagonal and off-diagonal disorders, **Chin. Phys. B** **22**, 127201 (2013)
12. **Limin Wang**, T. Berlijn, Y. Wang, C.-H. Lin, P. J. Hirschfeld, and W. Ku, *Effects of Disordered Ru Substitution in BaFe<sub>2</sub>As<sub>2</sub>: Possible Realization of Superdiffusion in Real Materials*, **Physical Review Letters** **110**, 037001 (2013).
13. K. F. Wang, **Limin Wang** and C. Petrovic, *Giant magnetothermopower effect in Dirac materials (Sr/Ca)MnBi<sub>2</sub>*, **Applied Physics Letters** **100**, 112111 (2012).
14. C.-H. Lin, T. Berlijn, **Limin Wang**, C. C. Lee, W. G. Yin, and W. Ku, *One-Fe versus Two-Fe Brillouin Zone of Fe-Based Superconductors: Creation of the Electron Pockets by Translational Symmetry Breaking*, **Physical Review Letters** **107**, 257001 (2011).
15. **Limin Wang** and Weiyi Zhang, *Disorder-type-dependent phase diagram in a CsCl-type complex Anderson lattice Model*, **Physical Review B** **81**, 054204 (2010).
16. **Limin Wang**, Weiyi Zhang, and Zhenxing Liu, *Properties of a unique type of critical state in two-dimensional two-band Anderson lattice model in the presence of site-selective disorder*, **Journal of the Physical Society of Japan**, **77**, 124707 (2008).
17. **Limin Wang**, Zhenxing Liu, and Weiyi Zhang, *Localization and scaling behavior of polariton-gap in Fibonacci and symmetric Fibonacci-class superlattices*, **Journal of the Physical Society of Japan** **77**, 034701 (2008).